Differentiate your Objective

AICE3002 Introduction to Deep Learning
AICE6001 Introduction to Deep Learning (MSc)

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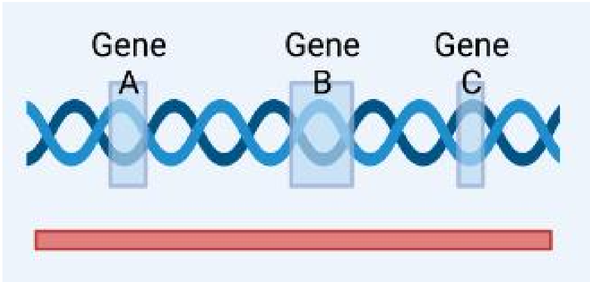
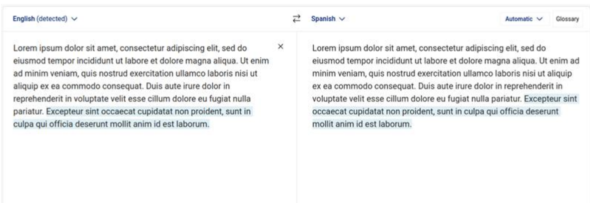
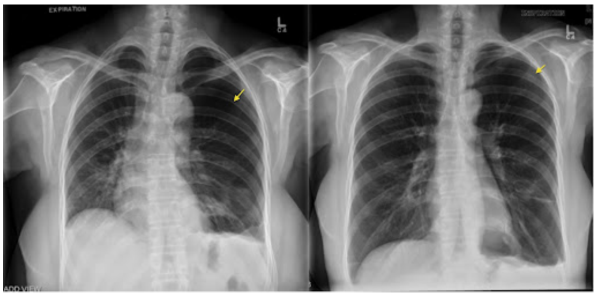
What does Deep Learning (and Differentiable Programming) give us?







Machine Learning - A Recap

All credit for this slide goes to Niranjan

Data	$\{\boldsymbol{x}_n, \boldsymbol{y}_n\}_{n=1}^N \qquad \{\boldsymbol{x}_n\}_{n=1}^N$
Function Approximator	$\boldsymbol{y} = f(\boldsymbol{x}; \boldsymbol{\theta})$
Parameter Estimation	$E_0 = \sum_{n=1}^N \{\|\boldsymbol{y}_n - f(\boldsymbol{x}_n; \boldsymbol{\theta})\|\}^2$
Prediction	$\hat{\boldsymbol{y}}_{N+1} = f(\boldsymbol{x}_{N+1}, \hat{\boldsymbol{\theta}})$

What is Deep Learning?

Deep learning is primarily characterised by function compositions:

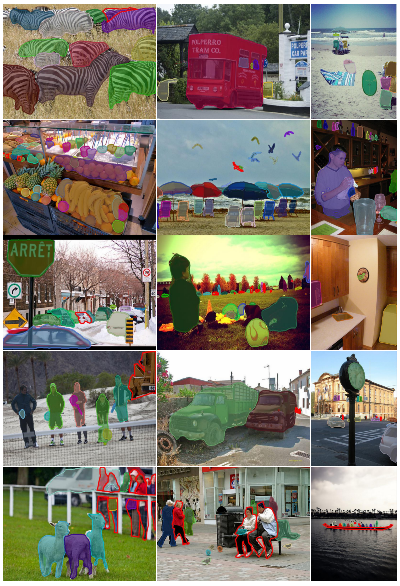
- Feedforward networks: $\boldsymbol{y} = f(g(\boldsymbol{x}; \boldsymbol{\theta}_g); \boldsymbol{\theta}_{\boldsymbol{f}})$
  - Often with relatively simple functions (e.g. $f(\boldsymbol{x}; \boldsymbol{\theta}_f) = \sigma(\boldsymbol{x}^\top \boldsymbol{\theta}_f)$)
- Recurrent networks:
  $\boldsymbol{y}_t = f(\boldsymbol{y}_{t-1}, \boldsymbol{x}_t; \boldsymbol{\theta}) = f(f(\boldsymbol{y}_{t-2}, \boldsymbol{x}_{t-1}; \boldsymbol{\theta}), \boldsymbol{x}_t; \boldsymbol{\theta}) = \dots$

What is Differentiable Programming?

- Captures the idea that computer programs can be partially learned rather than fully programmed by humans.
- The program is made of functions with optimisable parameters.
- So we need to be able to compute gradients with respect to the parameters of these functional blocks.
- The functional blocks don't need to be direct functions in a mathematical sense; more generally they can be *algorithms*.
- What if the functional block we're learning parameters for is itself an algorithm that optimises the parameters of an internal algorithm using a gradient based optimiser?![1]

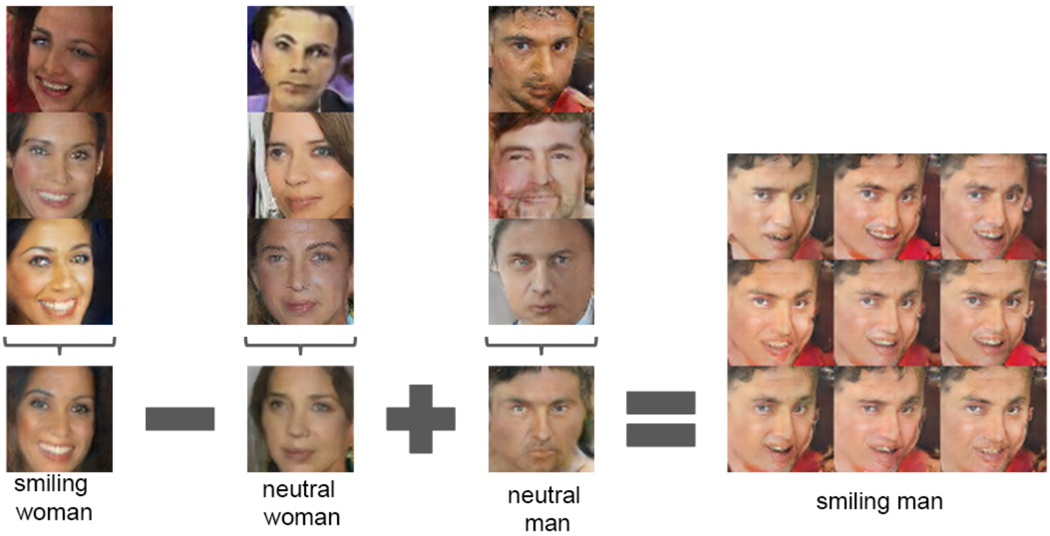
[1] See our ICLR 2019 paper: https://arxiv.org/abs/1812.03928 and NeurIPS 2019 paper: https://arxiv.org/abs/1906.06565

Success stories - Object detection and segmentation



Pinheiro, Pedro O., et al. "Learning to refine object segments." European Conference on Computer Vision. Springer, 2016.

Success stories - Image generation



Radford, Alec, Luke Metz, and Soumith Chintala. "Unsupervised representation learning with deep convolutional generative adversarial networks." arXiv preprint arXiv:1511.06434 (2015).

Success stories - Translation

GPT-4

Prompt:
Translate the following Irish text to English:
"Bhí mé ag siúl sa choill nuair a chuala mé ceol álainn ón adharc."

Output (With LLM Integration):
I was walking in the forest when I heard beautiful music from the horn.
Output (Traditional MT):
I was in the wood when I heard nice sound from the horn.

Lyu, C., Du, Z., Xu, J., Duan, Y., Wu, M., Lynn, T., Aji, A.F., Wong, D.F., Liu, S. and Wang, L., 2023. A Paradigm Shift: The Future of Machine Translation Lies with Large Language Models. LREC-COLING 2024

Why should we care about this?

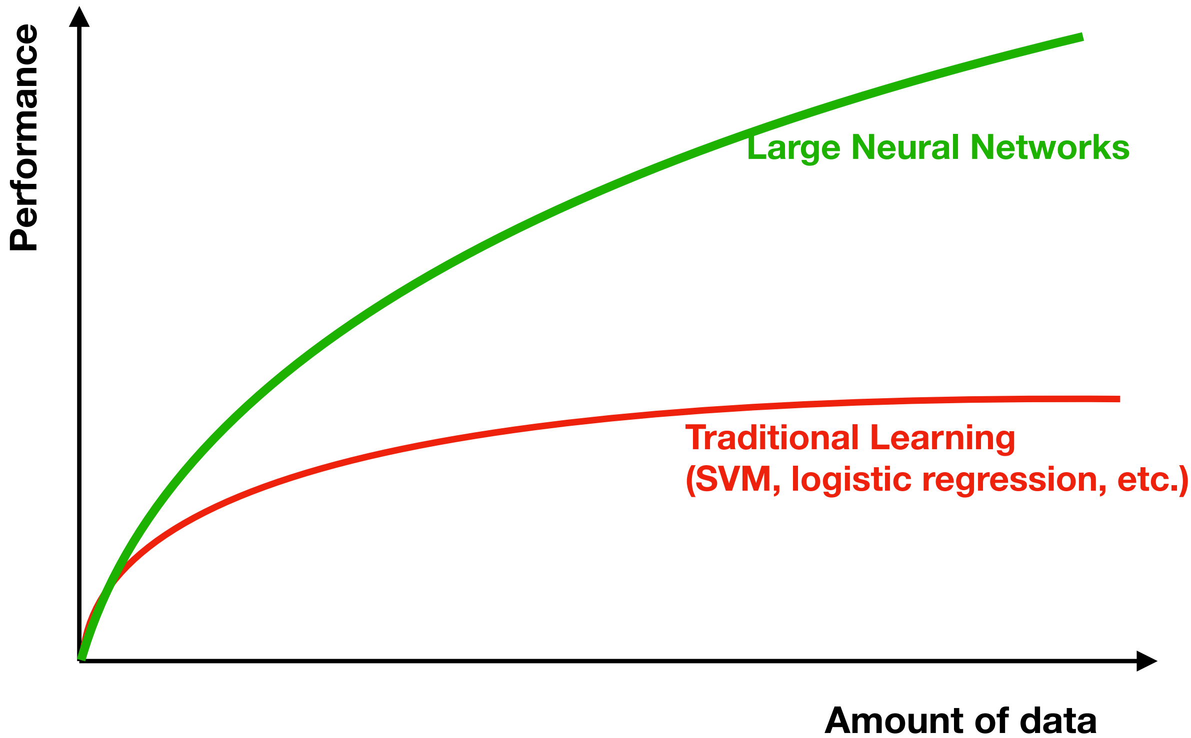


Image courtesy of Andrew Ng

Key takeaways

- **More data availability** in all domains...
- ...allows for **bigger/deeper/more complex models**
- ...models that can **learn the features**
- and these can have far **greater performance**

What is the objective of this module?

A word of warning: This is not a module about how to apply someone else's deep network architecture to a task, or how to train existing models!

You will learn some of that along the way of course, but the real objective is for you to graduate understanding how deep learning works beneath the hood.

What is the objective of this module?

- Understand the underlying mathematical and algorithmic principles of deep learning.
- Apply deep learning methods to real datasets.
- Learn how to train models using deep learning libraries.
- Think critically about why certain approaches were designed in the way they were.

How is this module going to be delivered?

- Lectures (3 per week except when we have seminars)
  - Note: We are refreshing some material from last year, but the website may have old links.
  - You need to read the suggested papers/links before the lectures!
  - There is maybe a little room for some flexibility later in the course on topics - tell us what you're interested in!
  - Lectures will be face to face, but also recorded for the website when possible (do not rely on this!).

How is this module going to be delivered?

- Seminars (4)
  - We'll look at a particular paper, papers or ideas in some detail and have an open discussion
  - You'll need to prepare in advance & be ready to ask questions and share your thoughts
  - Not recorded, but contents examinable in the quizzes

How is this module going to be delivered?

- Labs (1x 2 hour session per week for 8 weeks + additional help sessions if required)
  - Labs consist of a number of Juypter notebooks you will work though.
    - You'll be using PyTorch as the primary framework, with Torchbearer to help out.
    - You will need to utilise GPU-compute for the later labs (we provide Google Colab links so you can use NVidia K80s or newer in the cloud).
  - Labs are in-person (AICE lab in B32) with a team of PhD student demonstrators & both of us.
  - Please ask lots of questions and use this time to get help on the labs and coursework.
  - After each lab you will have to do a follow-up problem-sheet exercise that will be marked.

What will we cover in the module?

https://moodle.ecs.soton.ac.uk/course/view.php?id=469

Lab session plan

Lab	Date	Topic
Lab 1	3/02/25	Introducing PyTorch
Lab 2	10/02/25	Automatic Differentiation
Lab 3	17/02/25	Optimisation
Lab 4	24/02/25	NNs with PyTorch and Torchbearer
Lab 5	03/03/25	CNNs with PyTorch and Torchbearer
Lab 6	10/03/25	Transfer Learning
Lab 7	17/03/25	RNNs, Sequence Prediction and Embeddings
Lab 8	21/04/25	Deep Generative Models

What do we expect you already know?

- You have have taken Foundations of Machine Learning, or are currently taking it.
- Fundamentals of:
  - Matrix Algebra (matrix-matrix, matrix-vector and matrix-scalar operations, inverse, determinant, rank, Eigendecomposition, SVD);
  - Probability & Statistics (1st-order summary statistics, simple continuous and discrete probability distributions, expected values, etc); and,
  - Multivariable Calculus (partial differentiation, chain-rule).
- Programming in Python

What might you already know?

- How to use a deep learning framework (Keras, Tensorflow, PyTorch)?
- How to train an existing model architecture using a GPU?
- How to perform transfer learning?
- How to perform differentiable sampling of a Multivariate Normal Distribution?

Assessment Structure

- Lab work 40% - Handin in week 11 (6th May, 4PM)
- Final exam 60%

What comes after this module?

- AICE6002: Deep Learning Research
  - taught by both of us
  - focus on translating the fundamentals you learn in this module to cutting edge research